

بسم الله الرحمن الرحيم

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Ministry of Higher Education  
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**MATH329 : Operations Research**

**MidTerm Exam Mon. 13/04/2026**

**Time: 60 Min.**

**الرقم الجامعي:**

**اسم الطالب :**

**Question 1 : Answer true or false for all of the following:**

**( 7 M )**

1. Graphical solutions are only applicable to LP problems with more than three variables. F
2. Operations Research was first initiated during World War I in England to optimize war material utilization.
3. In a linear programming problem, the objective function or the constraints must be linear? F
4. The Simplex tableau is used to organize the objective function and constraint equations for the algorithm. T
5. The solution of the LPP may be compatible with its constraints imposed by the available resources. F
6. The Simplex Algorithm requires the coefficients of the starting basic variables to form an identity matrix. T
7. In Simplex method, if no variable can enter the basis, the problem has no feasible solution. T

**Question 2: Solve the following problem Graphically:**

**( 8 M )**

$$\text{Maximum } Z = 5x_1 + 3x_2$$

*S.to:*

$$2x_1 + 3x_2 \leq 12$$

$$2x_1 - x_2 \leq 8$$

$$\text{Maximum } Z = 5x_1 + 3x_2$$

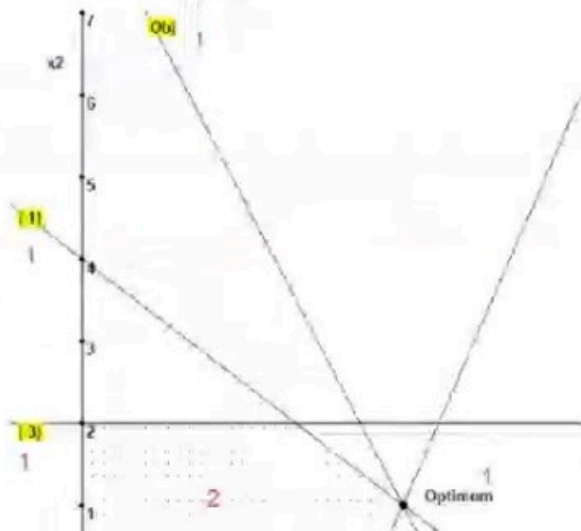
S.to:

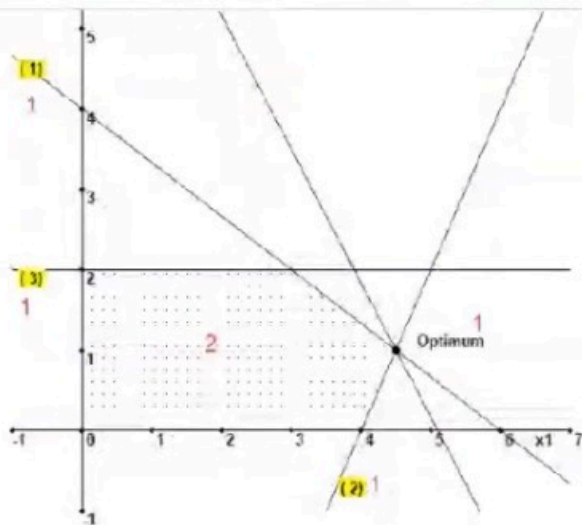
$$2x_1 + 3x_2 \leq 12$$

$$2x_1 - x_2 \leq 8$$

$$-2x_2 \geq -4$$

$$x_1 \geq 0, x_2 \geq 0$$





Objective Value = 25.5 at  $x_1 = 4.5$  and  $x_2 = 1$       2

Question 3: Given the following problem:

(5 + 10 M)

$$\text{Maximize } z = 16x_1 + 15x_2$$

Subject to:

$$40x_1 + 31x_2 \leq 124$$

$$x_1 - x_2 \geq -1$$

$$x_1 \leq 3$$

$$x_1, x_2 \geq 0$$

- a) Write the problem in **standard form**. 5  
b) Solve the problem using *Simplex Algorithm* (Stop at first Iteration "2 tables only"). 10

a) We adding slack variables, the problem becomes:

$$\text{Maximize } Z = 16x_1 + 15x_2 + 0s_1 + 0s_2 + 0s_3$$

$$\text{Subject to: } 40x_1 + 31x_2 + s_1 = 124$$

$$-x_1 + x_2 + s_2 = 1$$

$$x_1 + s_3 = 3$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

b) We start the solution by writing the Simplex tableau.

a) We adding slack variables, the problem becomes:

$$\text{Maximize } Z = 16x_1 + 15x_2 + 0s_1 + 0s_2 + 0s_3$$

Subject to:

$$40x_1 + 31x_2 + s_1 = 124$$

$$-x_1 + x_2 + s_2 = 1$$

$$x_1 + s_3 = 3$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

b) We start the solution by writing the Simplex tableau.

| Basic | x1  | x2  | s1 | s2 | s3  | Sol. |
|-------|-----|-----|----|----|-----|------|
| Z     | -16 | -15 | 0  | 0  | 0   | 0    |
| s1    | 40  | 31  | 1  | 0  | 0   | 124  |
| s2    | -1  | 1   | 0  | 1  | 0   | 1    |
| s3    | 1   | 0   | 0  | 0  | 1   | 3    |
| Z     | 0   | -15 | 0  | 0  | 16  | 48   |
| s1    | 0   | 31  | 1  | 0  | -40 | 4    |
| s2    | 0   | 1   | 0  | 1  | 1   | 4    |
| x1    | 1   | 0   | 0  | 0  | 1   | 3    |

(Every table 5 m and . Every mistake in calculation - 0.5m)